

# Chemical Kinetics: Formula Sheet

## Class 12 Chemistry – Unit 3 Complete Revision Formulas

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### 1 Rate of Reaction & Stoichiometry

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- **Average and Instantaneous Rate of Reaction:**

$$r_{\text{av}} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$
$$r_{\text{inst}} = -\frac{d[R]}{dt} = +\frac{d[P]}{dt} \quad [\text{Units: mol} \cdot \text{L}^{-1}\text{s}^{-1} \text{ or atm} \cdot \text{s}^{-1}]$$

- **General Stoichiometric Rate Expression ( $aA + bB \rightarrow cC + dD$ ):**

$$\text{Rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = +\frac{1}{c} \frac{d[C]}{dt} = +\frac{1}{d} \frac{d[D]}{dt}$$

- **Rate Law and Reaction Order:**

$$\text{Rate} = k[A]^x[B]^y \implies \text{Overall Order } n = x + y$$

Units of Rate Constant  $k$ :  $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$  (or  $\text{conc}^{1-n} \text{time}^{-1}$ ).

## 2 Integrated Rate Equations & Half-Life

- 1. Zero Order Reaction ( $n = 0$ ):

$$\text{Differential Rate: } -\frac{d[R]}{dt} = k$$

$$\text{Integrated Equation: } [R] = [R]_0 - kt \implies k = \frac{[R]_0 - [R]}{t}$$

$$\text{Half-Life Period: } t_{1/2} = \frac{[R]_0}{2k} \propto [R]_0$$

- 2. First Order Reaction ( $n = 1$ ):

$$\text{Differential Rate: } -\frac{d[R]}{dt} = k[R]$$

$$\text{Integrated Equation: } [R] = [R]_0 e^{-kt} \implies \ln[R] = \ln[R]_0 - kt$$

$$k = \frac{2.303}{t} \log \left( \frac{[R]_0}{[R]} \right) = \frac{2.303}{t_2 - t_1} \log \left( \frac{[R]_1}{[R]_2} \right)$$

$$\text{Half-Life Period: } t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k} \quad (\text{independent of } [R]_0)$$

- 3. First Order Gaseous Phase Reaction ( $A(g) \rightarrow B(g) + C(g)$ ):

$$k = \frac{2.303}{t} \log \left( \frac{p_i}{p_A} \right) = \frac{2.303}{t} \log \left( \frac{p_i}{2p_i - p_t} \right)$$

Where  $p_i$  is initial pressure of A, and  $p_t$  is total pressure at time  $t$ .

### 3 Temperature Dependence & Arrhenius Equation

- **Arrhenius Equation:**

$$k = A e^{-E_a/RT} \implies \ln k = \ln A - \frac{E_a}{RT} \implies \log k = \log A - \frac{E_a}{2.303 RT}$$

Where:

- $k$  = Rate constant of the reaction
- $A$  = Arrhenius frequency factor / pre-exponential factor
- $E_a$  = Activation energy [ $\text{J} \cdot \text{mol}^{-1}$ ]
- $R$  = Universal gas constant ( $8.314 \text{ J} \cdot \text{mol}^{-1}\text{K}^{-1}$ )
- $T$  = Absolute temperature in Kelvin [K]
- $e^{-E_a/RT}$  = Fraction of molecules with kinetic energy  $\geq E_a$

- **Temperature Variation Formula (Two Temperatures  $T_1$  and  $T_2$ ):**

$$\log \left( \frac{k_2}{k_1} \right) = \frac{E_a}{2.303 R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right] = \frac{E_a}{2.303 R} \left[ \frac{T_2 - T_1}{T_1 T_2} \right]$$

- **Linear Graph Parameters ( $\ln k$  vs  $1/T$ ):**

$$\text{Slope} = -\frac{E_a}{R}, \quad \text{Intercept} = \ln A$$

### 4 Collision Theory of Reaction Rates

- **Collision Theory Rate Equation:**

$$\text{Rate} = P Z_{AB} e^{-E_a/RT}$$

Where  $Z_{AB}$  is collision frequency between reactants  $A$  and  $B$ , and  $P$  is the steric / orientation probability factor.

- **Threshold Energy Relation:**

$$\text{Threshold Energy} = \text{Activation Energy } (E_a) + \text{Initial Energy of Reactants}$$